Bloque 2: Simulación de Carga de Trabajo y Monitorización

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1. Ejercicio 2.2.1 Carga Http con AB

```
juan@juan-HP-EliteBook-Folio-1040-G2:~$ ab -n 10000 -c 500 http://192.168.56.2/
This is ApacheBench, Version 2.3 <$Revision: 1903618 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking 192.168.56.2 (be patient)
Completed 1000 requests
Completed 2000 requests
Completed 3000 requests
Completed 4000 requests
Completed 5000 requests
Completed 6000 requests
Completed 7000 requests
Completed 8000 requests
Completed 9000 requests
Completed 10000 requests
Finished 10000 requests


Server Software:        Apache/2.4.62
Server Hostname:        192.168.56.2
Server Port:            80

Document Path:          /
Document Length:        73 bytes

Concurrency Level:      500
Time taken for tests:   8.005 seconds
Complete requests:      10000
```

Figura 1: Ejecucion sobre Apache_1.

```
Concurrency Level:      500
Time taken for tests:   8.005 seconds
Complete requests:      10000
Failed requests:        0
Total transferred:      3390000 bytes
HTML transferred:       730000 bytes
Requests per second:    1249.23 [#/sec] (mean)
Time per request:       400.246 [ms] (mean)
Time per request:       0.800 [ms] (mean, across all concurrent requests)
Transfer rate:          413.56 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    1   3.4      0      21
Processing:     8  391  59.9    411     539
Waiting:        1  390  61.3    411     539
Total:         23  392  56.9    411     551

Percentage of the requests served within a certain time (ms)
  50%    411
  66%    416
  75%    420
  80%    423
  90%    426
  95%    428
  98%    430
  99%    432
 100%    551 (longest request)
juan@juan-HP-EliteBook-Folio-1040-G2:~$
```

Figura 2: Ejecucion sobre Apache_2, continuación.

```
juan@juan-HP-EliteBook-Folio-1040-G2:~$ ab -n 10000 -c 500 http://192.168.56.3/
This is ApacheBench, Version 2.3 <$Revision: 1903618 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking 192.168.56.3 (be patient)
Completed 1000 requests
Completed 2000 requests
Completed 3000 requests
Completed 4000 requests
Completed 5000 requests
Completed 6000 requests
Completed 7000 requests
Completed 8000 requests
Completed 9000 requests
Completed 10000 requests
Finished 10000 requests


Server Software:        nginx/1.20.1
Server Hostname:        192.168.56.3
Server Port:            80

Document Path:          /
Document Length:        72 bytes

Concurrency Level:      500
Time taken for tests:   1.488 seconds
Complete requests:      10000
Failed requests:        0
```

Figura 3: Ejecucion sobre Nginx_1.

```
Concurrency Level:      500
Time taken for tests:   1.488 seconds
Complete requests:      10000
Failed requests:        0
Total transferred:      3030000 bytes
HTML transferred:       720000 bytes
Requests per second:    6719.06 [#/sec] (mean)
Time per request:       74.415 [ms] (mean)
Time per request:       0.149 [ms] (mean, across all concurrent requests)
Transfer rate:          1988.16 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        1   27  40.2     25    1074
Processing:    11   37  19.7     35     307
Waiting:        2   37  19.6     35     305
Total:         28   64  44.7     61    1102

Percentage of the requests served within a certain time (ms)
  50%     61
  66%     66
  75%     70
  80%     72
  90%     78
  95%     82
  98%     90
  99%     94
 100%   1102 (longest request)
juan@juan-HP-EliteBook-Folio-1040-G2:~$
```

Figura 4: Ejecucion sobre Nginx_2, continuación.

En esta prueba de carga se ha empleado la herramienta Apache Benchmark (ab) para someter a los servidores web Apache y Nginx a un test de estrés idéntico. Se ha definido un entorno de ejecución exigente , lanzando un total de 10.000 peticiones con un nivel de concurrencia de 500 peticiones simultáneas sobre el archivo estático index.html por defecto.
Ambos servidores han demostrado una configuración robusta ante la carga exigida, logrando despachar la totalidad de la prueba sin registrar ninguna petición fallida (Failed requests: 0).

Apache logró procesar una media de 1.249 peticiones por segundo (RPS), tardando aproximadamente 8 segundos en finalizar el test al completo. Su tiempo medio de respuesta por cada bloque de peticiones concurrentes fue de unos 400 milisegundos.
Nginx, por su parte, alcanzó una tasa de 6.719 peticiones por segundo (RPS), completando la misma carga en apenas 1,48 segundos. El tiempo medio de respuesta cayó drásticamente a unos 74 milisegundos. Nginx demostró ser más de 5 veces más rápido procesando este volumen.
Estas diferencias de rendimiento eran esperadas debido a la diferencia de arquitecturas entre ambos servidores.